

Business Process Technologies and Management

Exercises

juergen.mangler@tum.de

Assignment 4 - What to do (total 20 points)?

1. Export BPMN 2.0 XML from Signavio
2. Implement your own worklist implementation
 - `~user/worklist/orgmodel.xml` or `/ports/xxxxx/orgmodel.xml` - organisation model, has to include at least actors and roles.
 - `~user/worklist/add.php` or `/ports/xxxxx/add` - add a task to the worklist
 - `~user/worklist/ui.php` or `/ports/xxxxx/ui` - show the tasks of an identified actor. I.e. when going to the UI identification/login has to be possible.
 - Each actor can give back / refuse to do tasks
 - Each actor can do tasks, the results will be sent back to the CPEE.
 - Each task has a deadline after which the task has to be done by a different actor.
 - Implementation of task assignment algorithm - Divide your matrikel number by 3 and take the topic with the according remainder:
 - Actor can actively take tasks, and give them back - $\text{Matnr} \% 3 = 0$
 - Round Robin assign tasks, when actor is giving back a task, it is round robin assigned again - $\text{Matnr} \% 3 = 1$
 - Workload assign tasks, when actor is giving back a task it is assigned again, but not to the same actor - $\text{Matnr} \% 3 = 2$
 - 5 bonus points: find a mechanism to assign “similar” tasks (i.e. same customer) to the same actor.
3. Use cpee.org - <https://cpee.org/flow/>
 - Hook up the worklist to your user tasks. Parameters:
 - Organisation Model
 - Role
 - Deadline and/or reschedule info
 - Link to html with the form
 - Data elements to be filled into the form
4. 5 bonus points: implements SOD or BOD in your scenario. Do not hardcode it into the worklist or organisation model by define additional parameters to do it from the process.

Assignment 3 - Submission

Submission Artefacts:

1. Export BPMN 2.0 XML from Signavio - newest version
2. CPEE testset denoting the worklist
3. PDF with many screenshots showing the details of your solution.
4. Implementation (only source, no packages)

If your code is not hosted on lehre.bpm.in.tum.de, or does not work when worklist is started, 0 points will be awarded.

Deadline: as always, monday in 2 weeks, 23:59.

Please have first results in 1 week, so that we can discuss and answer questions in the course.

Assignment 3 - Things To Know

Assynchronous Interaction with CPEE works for all programming languages - you just need to send some HTTP headers. We propose the following minimal set of components in your implementation

Schedule:

- e.g. schedule.php; REST service
- receives all the parameters and stores them into a list (table in sqlite, or json file)
- POST parameters:
 - process context (data elements) that are sent from the process engine
 - which UI should be used
 - which participant role should be used
- POST headers:
 - a url to contact cpee.org -> to send back the data later: it is stored in the CPEE-CALLBACK header (Cpee_Callback in PHP)
- it MUST return the CPEE-CALLBACK: true HTTP response header to tell the engine to wait for a callback.

List:

- e.g. list.php; UI exclusively used by a process participant
- displays all scheduled work requests FOR ONE PARTICULAR PARTICIPANT
- upon click on one item change to ...

Item:

- e.g. item.php; UI exclusively used by a process participant
- displays all details for particular work item: (1) custom UI, (2) submitted process context (data elements)
- allows the user to submit new information in a particular format to ...

Callback:

- e.g. callback.php; REST service that interact with cpee.org
- sends all data provided by the process participant in “**Item**” back to the CPEE (stored in your SQLITE/file; see above)
- redirect to “**Schedule**”